Economics S5040: Dynamic Stochastic Modeling
Lecture 13 — Optimal Consumption and Saving

Eric Fisher

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1 The Transition from Solow

Last week,

$$I_t = sY_t,$$

and s was given. This week:

$$\boxed{\text{Why do people save?}}$$

Saving becomes a household choice rather than a parameter.

2 Cake Eating

A household begins period t with resources x_t and chooses

$$0 \leq c_t \leq x_t.$$

Unconsumed resources become tomorrow's state:

$$\boxed{x_{t+1} = x_t - c_t.}$$

The objective is

$$\max_{\{c_t\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

The state is x_t; the control is c_t; saving is $a_t = x_t - c_t$.

3 Bellman Equation

Let $V(x)$ be maximum lifetime utility from state x:

$$\boxed{V(x) = \max_{0 \leq c \leq x} \{u(c) + \beta V(x - c)\}.}$$

Current consumption gives $u(c)$; saving creates continuation value $\beta V(x - c)$. Thus

$$\boxed{\text{saving sacrifices current utility for future value.}}$$

The consumption policy is $c^*(x)$ and the saving policy is

$$a^*(x) = x - c^*(x).$$

4 First-Order and Envelope Conditions

For an interior optimum,

$$u'(c_t) = \beta V'(x_t - c_t) = \beta V'(x_{t+1}).$$

The envelope condition gives

$$V'(x_t) = u'(c_t).$$

Applying this at $t+1$ yields the Euler equation

$$\boxed{u'(c_t) = \beta u'(c_{t+1}).}$$

The marginal utility lost by consuming one unit less today equals the discounted marginal utility obtained tomorrow.

5 CRRA Preferences

Use

$$u(c) = \begin{cases} \dfrac{c^{1-\rho} - 1}{1-\rho}, & \rho \neq 1, \\ \log c, & \rho = 1. \end{cases}$$

Then $u'(c) = c^{-\rho}$ and

$$c_t^{-\rho} = \beta c_{t+1}^{-\rho}.$$

Hence

$$\boxed{\frac{c_{t+1}}{c_t} = \beta^{1/\rho}.}$$

Because $\beta < 1$, consumption declines over time: the cake has gross return one, while future utility is discounted.

6 The Exact Policy

Let $q = \beta^{1/\rho}$. Then $c_{t+j} = q^j c_t$. Since the entire cake is eventually consumed,

$$x_t = \sum_{j=0}^{\infty} c_{t+j} = c_t \sum_{j=0}^{\infty} q^j = \frac{c_t}{1-q}.$$

Therefore

$$\boxed{c^*(x) = (1 - \beta^{1/\rho})x}$$

and

$$\boxed{a^*(x) = \beta^{1/\rho} x.}$$

For log utility,

$$\boxed{c^*(x) = (1-\beta)x, \qquad a^*(x) = \beta x.}$$

7 Comparative Statics

A higher β makes the household more patient, reducing current consumption and increasing saving. Curvature ρ determines willingness to shift consumption across dates. The Euler equation summarizes both effects.

For example, with

$$x_0 = 100, \qquad \beta = .95, \qquad \rho = 2,$$
$$q = \sqrt{.95} \approx .9747.$$

The household consumes only a small fraction initially and smooths the cake over many periods.

8 Why This Simple Problem Matters

Cake eating has one state, one continuous control, no income, no production, and no uncertainty. This lets us isolate the central mechanism:

consumption today versus continuation value tomorrow.

More realistic models change the transition law, but not this logic.

9 From Exact to Numerical Solutions

Here we know

$$c^*(x) = (1 - \beta^{1/\rho})x.$$

Most dynamic programs have no closed form. Next lecture we will pretend we do not know this answer and recover it from the Bellman equation using:

1. a grid for the state;
2. a guess for V;
3. Bellman maximization;
4. value-function iteration;
5. interpolation;
6. recovery of the policy function.

The exact solution will let us measure numerical error.

10 What to Remember

1. Solow takes saving as given; optimal saving makes it a choice.
2. State: x; control: c; saving: $x - c$.
3. $$V(x) = \max_{0 \le c \le x} \{u(c) + \beta V(x - c)\}.$$

4.

$$u'(c_t) = \beta u'(c_{t+1}).$$

5. Under CRRA,

$$c^*(x) = (1 - \beta^{1/\rho})x.$$

6. The analytical solution provides a benchmark for numerical methods.